

Intermediate_Elective_2

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2026-05-15

1. Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)
library(NatParksPalettes)

# reading .xlsx files
library(readxl)

# wrap axis labels
library(scales)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)
```

Data

```

# read in vegetation data
veg <- read_csv(here("data", "veg.csv"))

# vegetation survey metadata
metadata <- read_csv(here("data", "vp_veg_metadata.csv"))

# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  # powerful argument in read_xlsx to show the missing variables as numbers
  na = c("", "n/a", "N/A", "over", "173+"))

```